[D.3.1] Universal AI: Solomonoff Induction

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The big picture

- **Universal AI** asks: what would an *optimally intelligent* agent look like, given unlimited compute and the weakest possible assumptions?
- It splits into two problems:
  1. **Prediction (Solomonoff):** passively observe a sequence $x_1, x_2, \ldots$ and predict the next symbol. *How should you learn?*
  2. **Action (AIXI):** interact with an unknown environment, take actions, collect reward. *How should you act?*
- Constructed as a single **Bayesian mixture** ξ over a class $\mathcal{M}$ of candidate environments, weighted by a prior w_ν.
  - Solomonoff: ξ (eventually) predicts as well as the truth μ.
  - AIXI: the Bayes-optimal policy π^*_ξ (eventually) acts as well as the optimal policy π^*_μ for true environment μ.
- Choose $\mathcal{M} =$ all computable environments, prior $w_\nu = 2^{-K(\nu)}$ (K is the "complexity" of the environment). Assumption "$\mu \in \mathcal{M}$" becomes "*the universe is computable*" and the prior $w_\nu = 2^{-K(\nu)}$ encodes *Occam's razor* (simpler universes are more likely).

Formalization

- A true (unknown) environment μ generates a binary sequence; each bit $x_t \sim \mu(\cdot \mid x_{<t})$.
- A **predictor** P sees the past $x_{<t}$ and outputs a distribution $P(\cdot \mid x_{<t})$ over the next bit (next-token prediction)
- **Quality measure:** Cumulative μ-expected squared prediction error $S_\infty^\mu := \sum_{t=1}^\infty s_t^\mu$

Per-step squared prediction error s_t^μ

$$s_t^\mu = \sum_{x_{<t}} \mu(x_{<t}) \left[\sum_{x_t} \left(P(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}) \right)^2 \right]$$

- **Goal:** build a P that makes cumulative error S_∞^μ *small*.
- **Idea:** don't bet on one model. Hedge your bets and mix over a whole class $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ of potential environments.

The Bayesian mixture ξ

Definition

Class $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ with prior weights $w_\nu > 0$, $\sum_\nu w_\nu = 1$.

$$\xi(x_{1:t}) := \sum_{\nu \in \mathcal{M}} w_\nu \, \nu(x_{1:t}).$$

- **Key fact (mixture is a predictor):** its one-step prediction is a *posterior-weighted* average of the members' predictions (you prove this today!)

$$\xi(x_t \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \, \nu(x_t \mid x_{<t}), \qquad w(\nu \mid x_{<t}) = w_\nu \, \frac{\nu(x_{<t})}{\xi(x_{<t})}.$$

- **Posterior update is multiplicative:** $w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \dfrac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}$: we reweight by how well ν predicted vs. ξ (Bayes' rule!)

Main result: the cumulative prediction bound

Cumulative bound

$$S_\infty^\mu := \sum_{t=1}^{\infty} \sum_{x_{<t}} \mu(x_{<t}) \sum_{x_t \in \mathbb{B}} \left(P(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}) \right)^2 \leq -\ln w_\mu$$

- Total squared error over *all time* is finite, bounded by the prior penalty on μ.
- Higher prior on the truth implies smaller total error.
- There are infinitely many environments. Cannot choose a uniform prior. What to choose?

Specializing to the Solomonoff prior

- Take the **Solomonoff prior** $w_\nu = 2^{-K(\nu)}$, where $K(\nu)$ is the *Kolmogorov complexity*: length of the shortest program computing ν.
- Substitute $w_\mu = 2^{-K(\mu)}$ into the cumulative bound:

Explicit complexity bound

$$S_\infty^\mu \leq K(\mu) \ln 2$$

- **Interpretation:** the total prediction error is (roughly) bounded by the **complexity** of the true environment. Simpler worlds are learned faster.
- This is the formal version of **Occam's razor**.

Other properties of the Solomonoff prior

- **Bounded mistakes:** on a deterministic sequence, predicting the most likely bit $\hat{x}_t = \arg\max_{x \in \mathbb{B}} \xi(x \mid x_{<t})$ makes at most $-2 \ln w_\mu$ wrong guesses *ever*.
- **Pareto optimality:** no other predictor beats ξ on *every* $\nu \in \mathcal{M}$ simultaneously (for KL loss *and* squared loss).
- **Misspecification ($\mu \notin \mathcal{M}$):** the bound degrades gracefully,

$$S_n^\mu \leq -\ln w_{\hat{\mu}} + D_n(\mu \parallel \hat{\mu})$$

where $\hat{\mu}$ is the closest model in $\mathcal{M}$ to μ.

Why care? Applications, approximations, critiques

- **A yardstick for prediction.** Solomonoff's bound [Sol78] is tight up to additive $\sqrt{\cdot}$ terms [Hut01] and Pareto-optimal for general losses [Hut03]; it dissolves classic puzzles of Bayesian confirmation (zero-prior, black ravens) [Hut07] and grounds a Bayesian theory of induction [RH11].
- **Computable cousins that work.** Shrink $\mathcal{M}$ and the same mixture becomes an algorithm: context tree weighting [WST95] is the exact Bayes mixture over all variable-order Markov models, in linear time; MDL [Grü07] replaces the mixture by its best single code; the speed prior [Sch02] trades universality for computability.
- **Prediction is compression.** The Hutter Prize scores intelligence by compressed size of Wikipedia; LLMs turn out to be strong general-purpose compressors [DRD+24].
- **Deep learning through this lens.** Meta-training nets on UTM-generated data amortises Solomonoff induction [GMGH+24]; transformers conjectured as its best extant approximation [YW25]; K-complexity explains the simplicity bias behind generalisation [GFRW24, MRVPL25].
- **The critiques.** No elegant *computable* universal predictor exists [Leg06]; different UTMs give non-equivalent priors [WSH13]; the "Occam's razor" reading is contested: the real assumption is computability of the world [Ste16, Ste26].

Takeaways

- **Solomonoff** solves induction: mix over all computable hypotheses, weighted by simplicity. Total prediction error $\leq K(\mu) \ln 2$.
- **AIXI** lifts this to action: the Bayes-optimal policy over the same universal mixture. It learns to predict *and* to act as well as if it knew the world.
- **The catch:** both are *incomputable* (K is incomputable; $\mathcal{M}$ is intractable to enumerate). They are gold standards, not computable algorithms, and the prior's $2^{-K(\nu)}$ hides a dependency on the choice of universal Turing machine.

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